Perspective-aware Healthcare answer summarization

Team JustInTime- Anika Roy, Radhikesh Agrawal, Talib Siddiqui

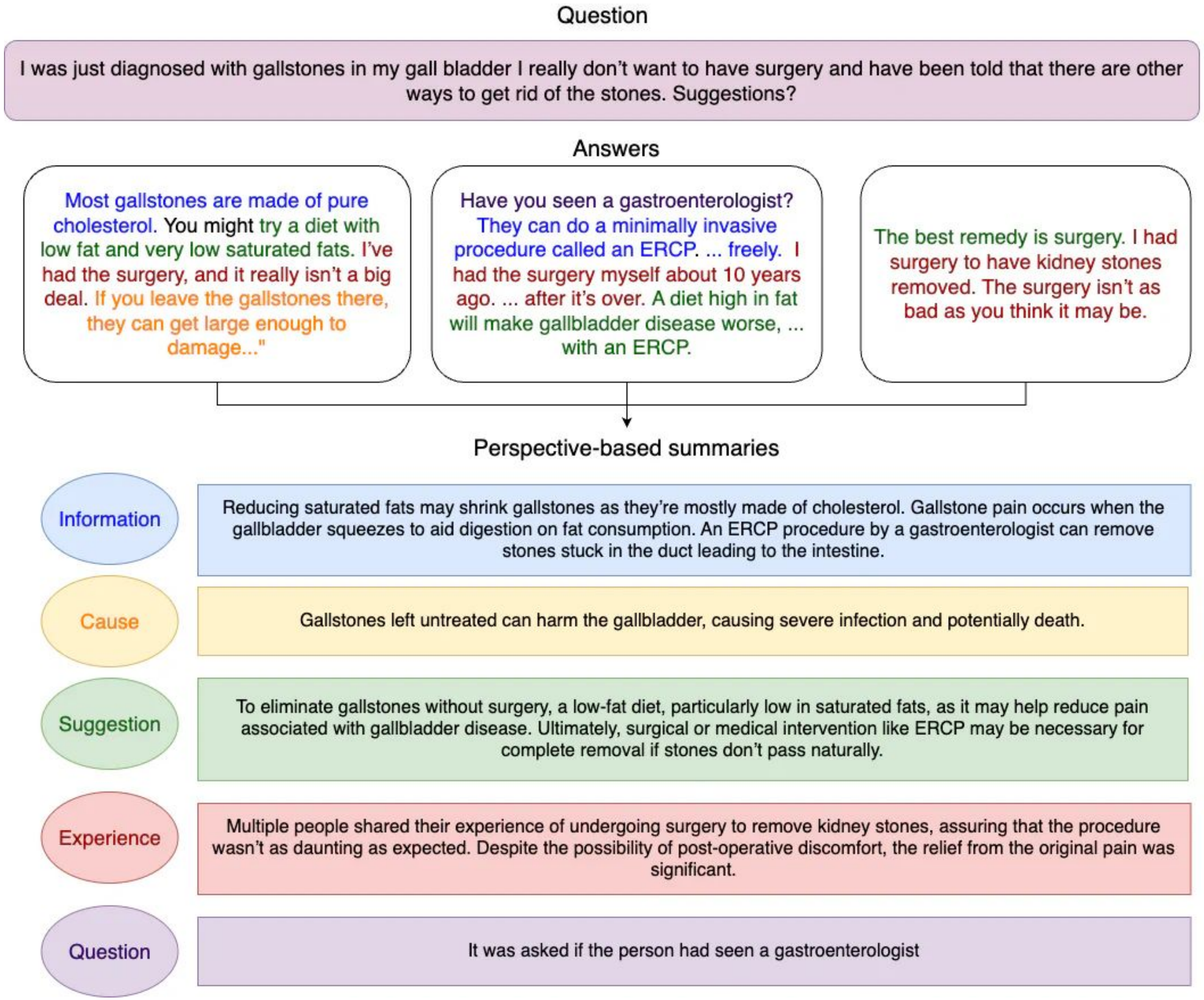
“A single answer often fails to capture the diverse perspectives presented across multiple answers.

Perspective specific summaries could better serve the information needs of end users.”

Core Objectives

- Healthcare Q&A often yields lengthy, diverse answers.
- Single answers can contain multiple viewpoints (e.g., causes, suggestions, experiences).
- Goal: Extract structured, perspective-specific information to improve user understanding.
  a. **Task A:** Identify & Classify Perspective Spans.
  b. **Task B**: Generate Perspective-Aware Summaries from the question-answer thread.

Example


Question
I was just diagnosed with gallstones in my gall bladder I really don't want to have surgery and have been told that there are other ways to get rid of the stones. Suggestions?
Answers
Most gallstones are made of pure cholesterol. You might try a diet with low fat and very low saturated fats. I've had the surgery, and it really isn't a big deal. If you leave the gallstones there, they can get large enough to damage..."
Have you seen a gastroenterologist? They can do a minimally invasive procedure called an ERCP. ... freely. I had the surgery myself about 10 years ago. ... after it's over. A diet high in fat will make gallbladder disease worse, ... with an ERCP.
The best remedy is surgery. I had surgery to have kidney stones removed. The surgery isn't as bad as you think it may be.
Perspective-based summaries
Information
Reducing saturated fats may shrink gallstones as they're mostly made of cholesterol. Gallstone pain occurs when the gallbladder squeezes to aid digestion on fat consumption. An ERCP procedure by a gastroenterologist can remove stones stuck in the duct leading to the intestine.
Cause
Gallstones left untreated can harm the gallbladder, causing severe infection and potentially death.
Suggestion
To eliminate gallstones without surgery, a low-fat diet, particularly low in saturated fats, as it may help reduce pain associated with gallbladder disease. Ultimately, surgical or medical intervention like ERCP may be necessary for complete removal if stones don't pass naturally.
Experience
Multiple people shared their experience of undergoing surgery to remove kidney stones, assuring that the procedure wasn't as daunting as expected. Despite the possibility of post-operative discomfort, the relief from the original pain was significant.
Question
It was asked if the person had seen a gastroenterologist

Understanding the Dataset

1. **uri:** Unique ID for the Q&A thread, e.g., "1504599".
2. **question:** The user's main query — "I scream, shout and swear in my sleep. How do I stop?"
3. **context:** Background info, e.g., user's history of sleep-talking and desire to stop.
4. **answers:** A list of user-generated replies, ranging from jokes ("duct tape") to serious advice (stress management, magnesium, see a doctor).
5. **labelled_answer_spans:** Annotated parts of each answer, categorized as:
   a. SUGGESTION (e.g., "try sleeping tablets")
   b. CAUSE (e.g., "stress in daily life")
   c. INFORMATION (e.g., "magnesium helps relax the body and mind")

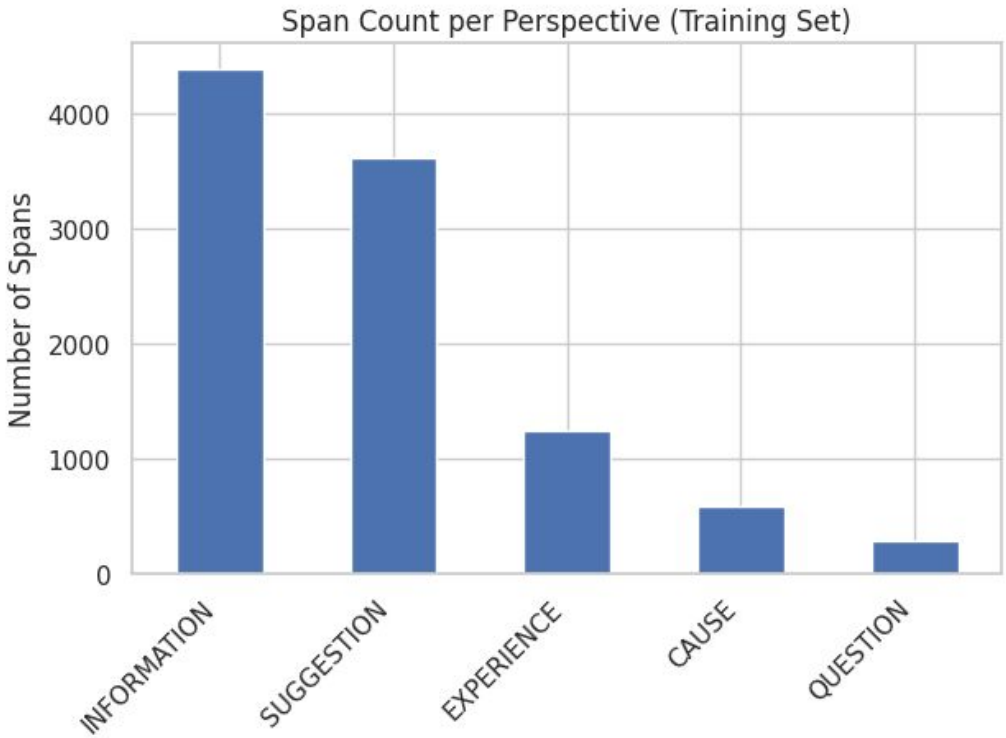
Understanding the Dataset

6. labelled_summaries: Condensed summaries for each category:

    a. CAUSE: Sleep issues may be stress-related
    b. INFORMATION: Magnesium aids restful sleep
    c. SUGGESTION: Broad advice like recording yourself, reducing stress, or seeing a psychiatrist

7. raw_text: The full text (question, context, answers), with labeled spans pointing to specific character offsets for traceability.

The PUMA Dataset

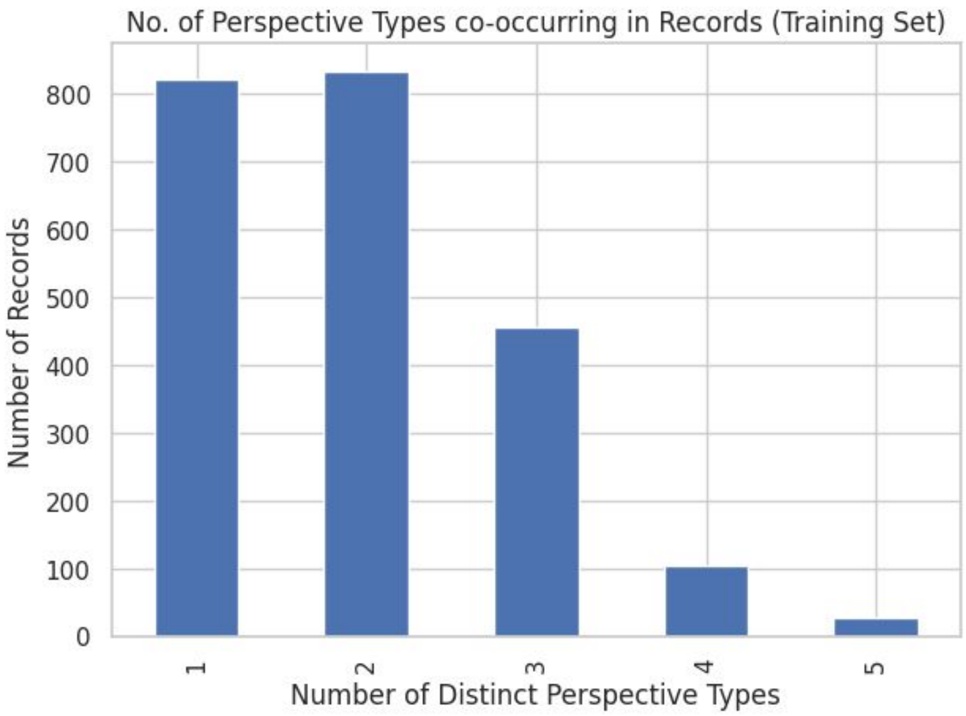


Perspective s**UM**marization d**A**taset

Uneven distribution of perspective spans

Information and **Suggestion** are most frequent

The PUMA Dataset



Uneven number of perspectives across records

1 and 2 perspective types per datapoint are most frequent

Task A: Approach - Finding the Perspectives

Method: Sequence Labeling using IOB2 scheme.

Model: BERT-base-uncased (*AutoModelForTokenClassification*).

Input: raw_text (Question + Context + Answers).

Output: Token-level labels (e.g., B-SUGGESTION, I-SUGGESTION, O).

Task A examples

```
Token: discharge        True: I-INFORMATION    Pred: I-INFORMATION
Token: "                True: I-INFORMATION    Pred: I-INFORMATION
Token: does             True: I-INFORMATION    Pred: I-INFORMATION
Token: not              True: I-INFORMATION    Pred: I-INFORMATION
Token: smell            True: I-INFORMATION    Pred: I-INFORMATION
Token: bad              True: I-INFORMATION    Pred: I-INFORMATION
Token: .                True: I-INFORMATION    Pred: I-INFORMATION
Token: is               True: I-INFORMATION    Pred: I-INFORMATION
Token: not              True: I-INFORMATION    Pred: I-INFORMATION
Token: gray             True: I-INFORMATION    Pred: I-INFORMATION
Token: ,                True: I-INFORMATION    Pred: I-INFORMATION
Token: green            True: I-INFORMATION    Pred: I-INFORMATION
Token: ,                True: I-INFORMATION    Pred: I-INFORMATION
Token: or               True: I-INFORMATION    Pred: I-INFORMATION
Token: yellow           True: I-INFORMATION    Pred: I-INFORMATION
```

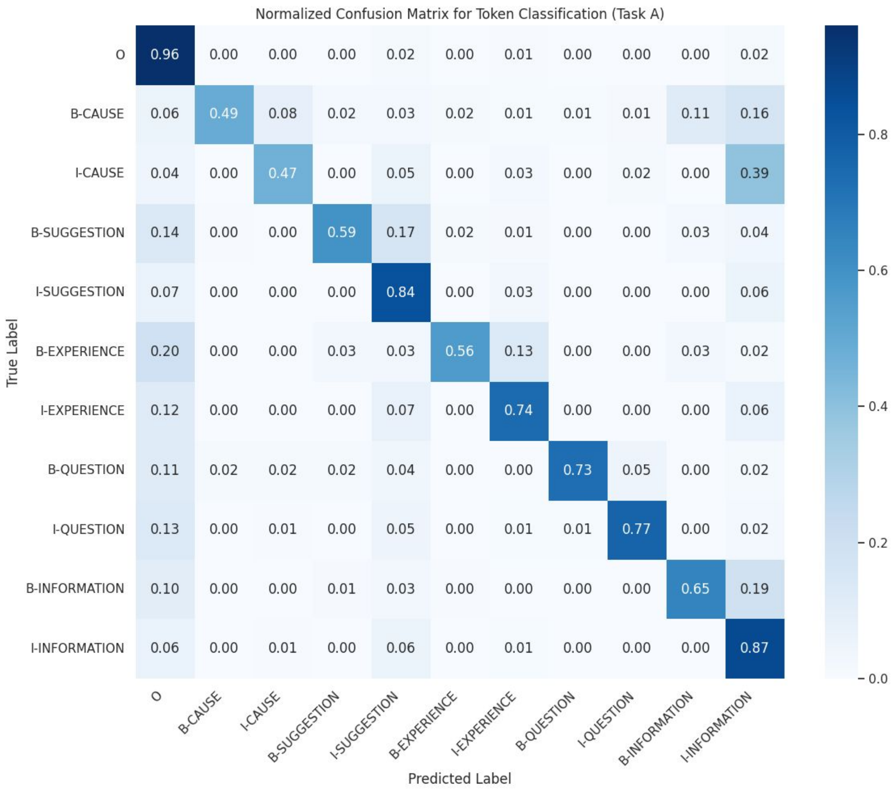
Task A examples

```
--- Misclassified Sample 2 (Original Index in Dataset might vary due to shuffle) ---
  Token: these          True: B-INFORMATION   Pred: B-EXPERIENCE    <<< MISMATCH
  Token: were           True: I-INFORMATION   Pred: I-EXPERIENCE    <<< MISMATCH
  Token: my             True: I-INFORMATION   Pred: I-EXPERIENCE    <<< MISMATCH
  Token: approximation  True: I-INFORMATION   Pred: I-EXPERIENCE    <<< MISMATCH
  Token: ##s            True: I-INFORMATION   Pred: I-EXPERIENCE    <<< MISMATCH
  Token: from           True: I-INFORMATION   Pred: I-EXPERIENCE    <<< MISMATCH
  Token: the            True: I-INFORMATION   Pred: I-EXPERIENCE    <<< MISMATCH
  Token: growth         True: I-INFORMATION   Pred: I-EXPERIENCE    <<< MISMATCH
  Token: charts         True: I-INFORMATION   Pred: I-EXPERIENCE    <<< MISMATCH
  Token: listed         True: I-INFORMATION   Pred: I-EXPERIENCE    <<< MISMATCH
  Token: below          True: I-INFORMATION   Pred: I-EXPERIENCE    <<< MISMATCH
  Token: :              True: I-INFORMATION   Pred: I-EXPERIENCE    <<< MISMATCH
  Token: 61             True: I-INFORMATION   Pred: I-EXPERIENCE    <<< MISMATCH
  Token: inches         True: I-INFORMATION   Pred: I-EXPERIENCE    <<< MISMATCH
  Token: (              True: I-INFORMATION   Pred: I-EXPERIENCE    <<< MISMATCH
  Token: 5              True: I-INFORMATION   Pred: I-EXPERIENCE    <<< MISMATCH
  Token: foot           True: I-INFORMATION   Pred: I-EXPERIENCE    <<< MISMATCH
  Token: 1              True: I-INFORMATION   Pred: I-EXPERIENCE    <<< MISMATCH
  Token: inch           True: I-INFORMATION   Pred: I-EXPERIENCE    <<< MISMATCH
  Token: )              True: I-INFORMATION   Pred: I-EXPERIENCE    <<< MISMATCH
  Token: and            True: I-INFORMATION   Pred: I-EXPERIENCE    <<< MISMATCH
  Token: 113            True: I-INFORMATION   Pred: I-EXPERIENCE    <<< MISMATCH
  Token: pounds         True: I-INFORMATION   Pred: I-EXPERIENCE    <<< MISMATCH
  Token: because        True: B-EXPERIENCE    Pred: I-EXPERIENCE    <<< MISMATCH
  Token: im             True: I-EXPERIENCE    Pred: B-EXPERIENCE    <<< MISMATCH
```

Task A examples

```
Token: iv            True: I-INFORMATION   Pred: I-INFORMATION
Token: ##not         True: B-SUGGESTION    Pred: I-INFORMATION    <<< MISMATCH
Token: ##hing        True: I-SUGGESTION    Pred: I-INFORMATION    <<< MISMATCH
Token: to            True: I-SUGGESTION    Pred: I-INFORMATION    <<< MISMATCH
Token: worry         True: I-SUGGESTION    Pred: I-INFORMATION    <<< MISMATCH
Token: about         True: I-SUGGESTION    Pred: I-INFORMATION    <<< MISMATCH
Token: at            True: I-SUGGESTION    Pred: I-INFORMATION    <<< MISMATCH
Token: all           True: I-SUGGESTION    Pred: I-INFORMATION    <<< MISMATCH
Token: yes           True: B-INFORMATION   Pred: B-INFORMATION
```

Task A: Results - How Well Did We Find Spans?



Semantic differentiation is the hardest part, especially for SUGGESTION vs. EXPERIENCE, and CAUSE vs. INFORMATION.

Performance for per token classification did well, but identification of correct contiguous spans was much harder

Task A: Results - How Well Did We Find Spans?

```
--- Seqeval Classification Report ---
              precision    recall  f1-score   support

       CAUSE       0.36      0.36      0.36       237
  EXPERIENCE       0.33      0.30      0.32       528
 INFORMATION       0.39      0.44      0.42      1661
    QUESTION       0.53      0.56      0.54       122
  SUGGESTION       0.40      0.38      0.39      1475

   micro avg       0.39      0.40      0.40      4023
   macro avg       0.40      0.41      0.41      4023
weighted avg       0.39      0.40      0.39      4023

Epoch 10: TrainLoss=0.0502, ValLoss=0.4393, ValF1=0.4064,
ValPrec=0.4042, ValRec=0.4096
```

Task A: Results - How Well Did We Find Spans?

Perspective	Precision	Recall	F1 Score	Support
CAUSE	0.36	0.36	0.36	237
EXPERIENCE	0.33	0.30	0.32	528
INFORMATION	0.39	0.44	0.42	1661
QUESTION	0.53	0.56	0.54	122
SUGGESTION	0.40	0.38	0.39	1475

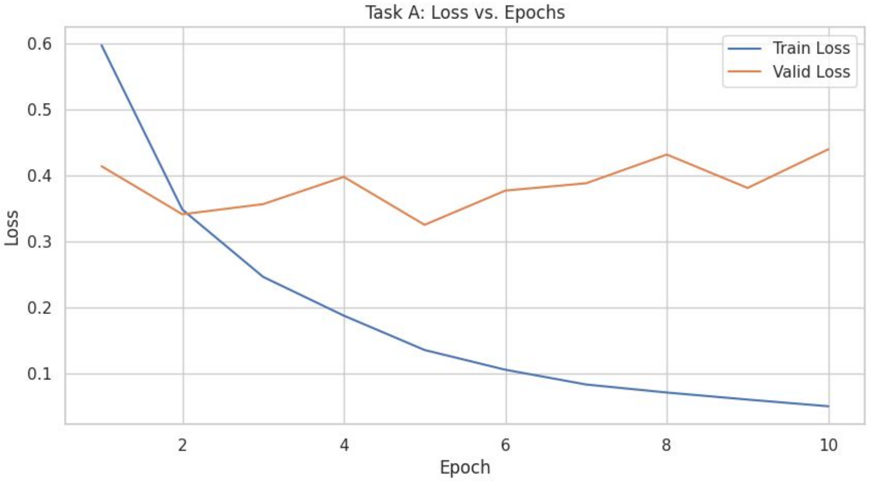
Task A: Results - How Well Did We Find Spans?

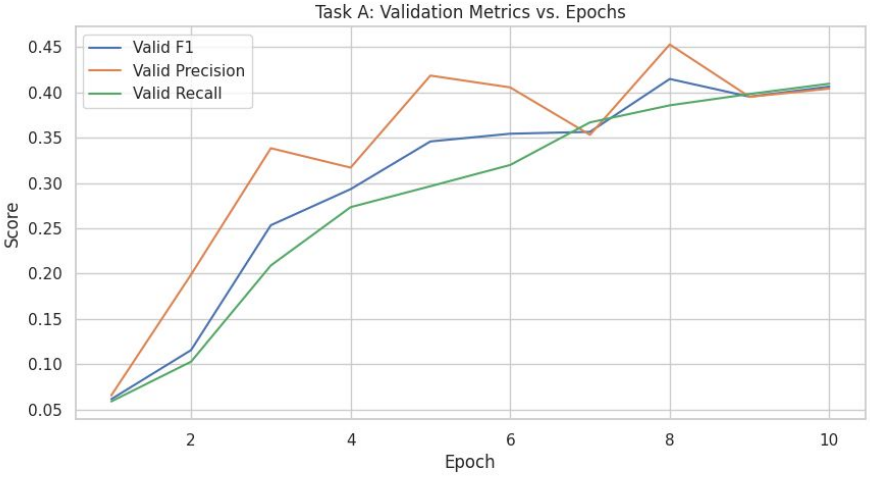
Average	Precision	Recall	F1 Score	Support
Micro Avg	0.39	0.40	0.40	4023
Macro Avg	0.40	0.41	0.41	4023
Weighted Avg	0.39	0.40	0.39	4023

Epoch 10: Best Validation F1= 0.4064 Train Loss = 0.0502 Val Loss = 0.4393

Explanation

- QUESTION perspective, despite having the lowest support (122 instances), achieved the highest F1-score (0.54). This suggests that 'QUESTION' spans might have distinct linguistic markers (e.g., question marks, interrogative words) that the model can learn effectively even with fewer examples.
- INFORMATION also performs relatively well (F1 of 0.42), benefiting from its high frequency in the dataset."
- Class Imbalance: The higher support for INFORMATION and SUGGESTION likely contributes to their reasonable F1 scores. The lower F1 for EXPERIENCE might still be linked to its moderate support but also potentially its linguistic diversity.
- **Task Complexity:** Strict span matching with IOB tags is a hard evaluation. The model has to get both the boundaries and the type exactly right."





The later plateau/increase in validation loss while F1 might still be high or slightly increasing is a classic sign that the model is memorizing training examples rather than improving its generalization ability.

Task B: Model Selection & Initial Prototyping

- **facebook/bart-base (BART)** – a seq2seq Transformer combining BERT's bidirectional encoder and GPT-style autoregressive decoder.
- Proven performance on summarization datasets and extensive available support
- *Alternative Considered*: Models like Flan-T5 offer potential benefits with instruction tuning but BART provides a strong starting point.

Task B: Fine-Tuning & Optimization

Baseline: Fine-tuned BART with simple prompts like:

"Summarize from perspective: SUGGESTION. Question: ... Answers: ..."

Prompt

Summarize from perspective: **INFORMATION**. Question: What are the causes and treatments for migraines? Answers: Migraines can be triggered by stress, hormonal changes, certain foods. Treatment includes pain relievers, preventive medications, and lifestyle changes.

Summary

For migraines, medication includes pain relievers, preventive medications, and lifestyle changes.

Task B: Fine-Tuning & Optimization

Prompt

Summarize from perspective: **CAUSE**. Question: What are the causes and treatments for migraines? Answers: Migraines can be triggered by stress, hormonal changes, certain foods. Treatment includes pain relievers, preventive medications, and lifestyle changes.

Summary

Migraines can be triggered by stress, hormonal changes, certain foods. Treatment includes pain relievers, preventive medications, and lifestyle changes.

Initial Versions of Task B

Key Observations:

- Model struggled to differentiate between perspectives with short or factual inputs.
- Prompts like "Summarize from perspective: X" were too weak on their own.
- Stronger template-based prompt engineering was needed.

Sample Results:

- Despite different perspective prompts (e.g., CAUSE, EXPERIENCE), summaries were often nearly identical.
- Shows the need for rich, varied inputs and better perspective conditioning.

Improved Approach

Prefix-tuning using PEFT (Parameter-Efficient Fine-Tuning) with structured prompts:

- Perspective definition
- Tone/style cues
- Start-of-summary phrases
- Full context (question + answers + user context)

Task B: Evaluation

We implemented a comprehensive evaluation pipeline for Task B using the following automatic metrics:

- ROUGE-1 F (Relevance)
- BLEU (N-gram overlap)
- BERTScore (Semantic similarity)

Each metric was computed:

- Per perspective (CAUSE, SUGGESTION, EXPERIENCE, INFORMATION, QUESTION)
- Macro-averaged across perspectives

Rich Input Examples

Example 1 (EXPERIENCE):

- Input: Long narrative about having braces.
- Reference Summary: Focused on pain during adjustments, adaptation over time, and role of age/treatment.
- Prediction: Captured a partial experience, but lacked depth.

Example 2 (SUGGESTION):

- Input: Depression-related suggestions (antidepressants, natural remedies, nutritionists).
- Prediction: Output was off-topic — hallucinated a weight loss suggestion → perspective drift.

Example 3 (INFORMATION):

- Input: Same as above, with educational tone.
- Prediction: Generated a weight-loss suggestion → topic and perspective mismatch.

Conditioning Strategy 2: Controlled Loss

Concept: Standard Cross-Entropy (I_{CE}) optimizes for likelihood, not explicit constraint satisfaction.
Our Approach: Augmented I_{CE} with a perspective-specific, energy-based loss term ($I_{perspective}$).

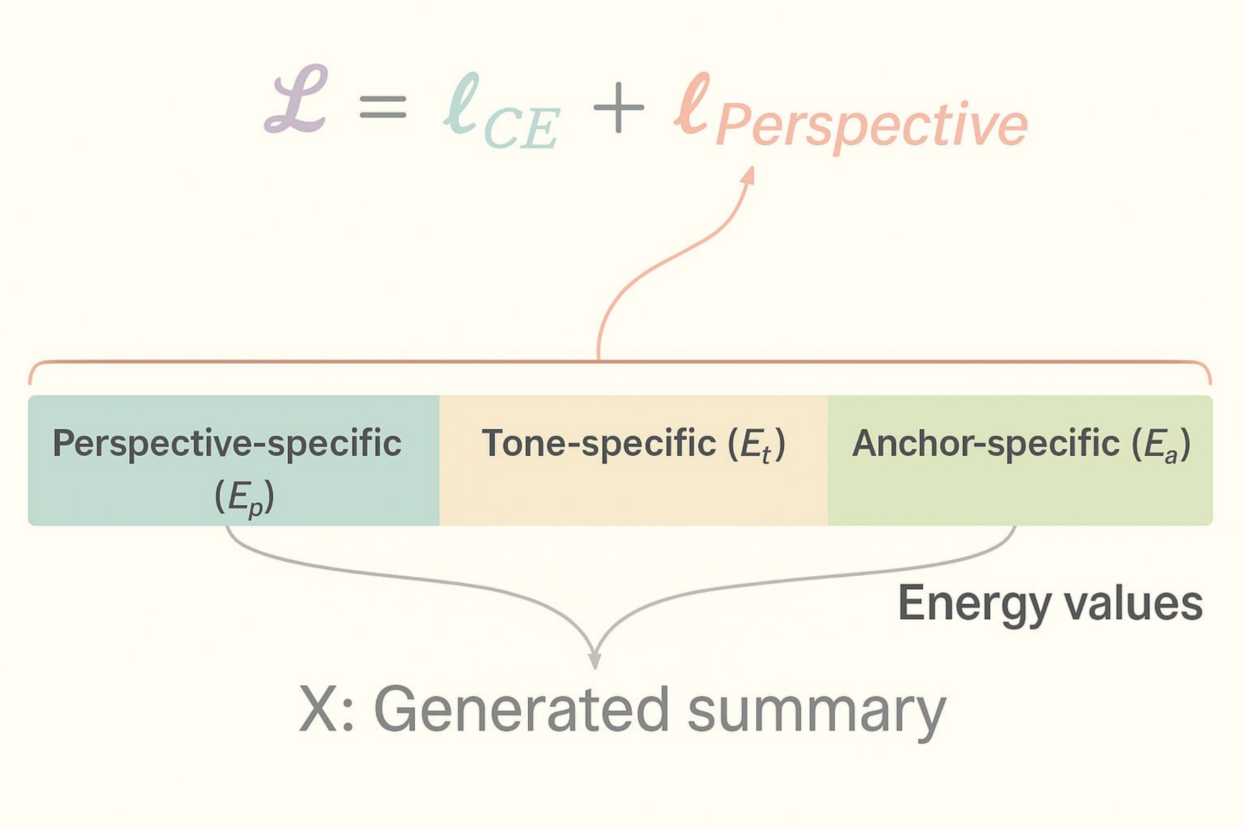
$$L_{total} = I_{CE} + \lambda * I_{perspective}$$ (λ balances the terms)

Energy Components (Conceptual): I_perspective derived from energy scores measuring:

- Ep: How well the generated summary matches the target Perspective type (e.g., using a classifier. We used roberta_tokenizer).
- Ea: How well the summary start matches the required Anchor text (We used ROUGE-1 on first few words).
- Et: How well the summary matches the desired Tone (We used cosine similarity of embeddings with tone keywords).

Goal: Penalize summaries that don't meet perspective conditions, guiding optimization beyond just likelihood.

Conditioning Strategy 2: Controlled Loss



$$p_i(S^P) = \frac{e^{-\frac{1}{E(S^P)_i}}}{\sum_j e^{-\frac{1}{E(S^P)_j}}}$$

$$\ell_{Perspective} = -\sum_i y_i \log(p_i(S^P))$$

Task B: Evaluation (normal loss)

Perspective	Count	ROUGE-1 F1	BLEU	BERTScore F1
EXPERIENCE	314	0.3580	0.1038	0.8855
INFORMATION	730	0.3541	0.0816	0.8890
SUGGESTION	593	0.3326	0.0859	0.8896
CAUSE	138	0.3234	0.0867	0.8847
QUESTION	101	0.4498	0.1952	0.9174
Macro-Average	—	0.3636	0.1106	0.8932

Task B: Evaluation (custom loss)

Perspective	Count	ROUGE-1 F1	BLEU	BERTScore F1
EXPERIENCE	314	0.3596	0.1087	0.8858
INFORMATION	730	0.3450	0.0741	0.8876
SUGGESTION	593	0.3327	0.0872	0.8899
CAUSE	138	0.3247	0.0864	0.8846
QUESTION	101	0.4359	0.1444	0.9044
Macro-Average	—	0.3596	0.1002	0.8905

Comparison and Reasoning

- **Complex Objective**: The model optimizes both fluency (Cross-Entropy) and constraint satisfaction (Ep, Ea, Et), creating a more complex optimization landscape.
- **Training Duration**: Three epochs may be insufficient for the model to fully adapt to the custom loss. Longer training is likely needed for benefits to emerge.
- **Lambda Sensitivity**: The λ (lambda) hyperparameter controls the weight of the custom loss:
  a. Too low: Minimal impact, especially early on.
  b. Too high: Over-penalizes fluency, hurting metrics like ROUGE and BLEU.

Per-Perspective Metrics

QUESTION

Highest in every metric — especially ROUGE and BERTScore.

It's possible that reference summaries for 'QUESTION' are very distinct and perhaps formulaic (e.g., 'It was asked if X...'), leading to better ROUGE/BLEU if the model learns this pattern. The high BERTScore suggests strong semantic capture of the inquiry.

Per-Perspective Metrics

INFORMATION

Consistently lowest across all metrics

While INFORMATION is often factual and should be easier, the very low BLEU suggests our model's summaries for information, while semantically similar (good BERTScore), are using quite different phrasing compared to the reference summaries.

Task B: Evaluation (normal loss) example

```
example_question = "What are the causes and treatments for Migraine?"
    example_answers = [
        "Migraines can be triggered by stress, hormonal changes, certain foods like aged cheese or chocolate, alcohol, sleep disturbances, and environmental factors such as bright lights or weather changes.",
        "Treatment for migraines involves acute strategies like over-the-counter pain relievers (ibuprofen) or prescription triptans. Preventive options include beta-blockers, anti-seizure drugs, antidepressants, and lifestyle changes like regular exercise and stress management. Identifying personal triggers is key."
    ]
```

Task B: Evaluation (normal loss) example

--- Perspective: INFORMATION ---
Generated Summary: For information purposes Migraines can be triggered by stress, hormonal changes, certain foods like aged cheese or chocolate, alcohol, sleep disturbances, and environmental factors like bright lights or weather changes. Treatment for migraines involves acute strategies like over-the-counter pain relievers (ibuprofen) or prescription triptans.

--- Perspective: SUGGESTION ---
Generated Summary: It is suggested For migraines, effective treatment involves over-the-counter pain relievers like ibuprofen or prescription triptans. Preventive options include beta-blockers, anti-seizure drugs, antidepressants, and lifestyle changes like regular exercise and stress management. Identifying personal triggers is crucial.

Task B: Evaluation (normal loss) example

--- Perspective: EXPERIENCE ---
Generated Summary: In user's experience In users experience, one individual has migraines due to stress, hormonal changes, certain foods like aged cheese or chocolate, alcohol, sleep disturbances, and environmental factors like bright lights or weather changes.

--- Perspective: CAUSE ---
Generated Summary: Some of the causes Migraines can be triggered by stress, hormonal changes, certain foods like aged cheese or chocolate, alcohol, sleep disturbances, and environmental factors like bright lights or weather changes.

--- Perspective: QUESTION ---
Generated Summary: It is inquired A follow-up question was asked if there were any personal triggers for Migraine.

Strong Examples

1) Medical Information (Accuracy):

- "Lisinopril is an ACE inhibitor that helps lower blood pressure, while Zocor is a statin..." (accurate differentiation of medications).

- "Nose hairs serve as a defense mechanism by filtering out harmful particles..." (clear biological explanation).

Strong Examples

2) Practical Advice (Suggestions):

- "To eliminate bad breath, flossing and brushing the back of the tongue is recommended." (concise and practical).

Weak Examples

1) **Experience (Lack of detail)**

Experience-based summaries, like anxiety treatment or bedwetting experiences, are often too generalized and lack personal specifics, same with diseases and conditions that do have constant symptoms throughout the lifetime of the condition; for eg. an autoimmune condition like colitis may not be same for the same person

Weak Examples

2) **Questions (Clarity)**

    Responses summarizing questions are sometimes unclear or overly vague

Future Directions

Fine-tuning on Questions and Anecdotes

Provide explicit training examples of questions and anecdotal experiences, this combined with prompt conditioning, can help capture nuances on QUESTION and EXPERIENCE

Generated "QUESTION" summary:
"It is inquired if it is blood or blood snot."

This summary is very vague and misses important context from the original detailed question and answer provided. It lacks clarity and context.
"What does it mean when you have the flu, blow your nose and all you get is blood?"

Fine-tuning on Questions and Anecdotes

Original Answers:
"Sometimes, when you have flu and blow your nose excessively, you might see blood. It could indicate dried sinuses, Vitamin C deficiency, or potentially something more serious."

—------

"The question seeks clarification on the potential causes of experiencing blood when blowing the nose during flu, with suggested reasons including dry sinuses, Vitamin C deficiency, or more serious conditions."

Fine-tuning on Questions and Anecdotes

Clearly states the context and intent of the original question.
Provides a structured overview of specific answers and possible causes mentioned in the discussion.

Retains essential detail and clarity, enhancing overall comprehensibility.

Prompt Conditioning

Improving instructions or prompts to emphasize specific details, particularly for "EXPERIENCE" and "QUESTION" perspectives."

For eg. here we have a generated "EXPERIENCE" summary:
"In users experience, one individual found out that drinking orange juice helps with their cold, another mentions that Zicam and Cold-eez really works."

This summary is overly vague and misses key experiential details from the original anecdotal answers.

Prompt Conditioning

If Prompt Improved, A more explicit prompt (e.g., "Summarize by clearly capturing personal anecdotes and specific details of experiences") could yield:

"In users' experiences, drinking orange juice significantly reduced symptoms, and products like Zicam and Cold-eez provided effective relief from colds, based on personal testimonies."

The improved prompt clearly instructs the model to retain experiential specifics and nuances.

Reduce Redundancy

Explicitly instruct the model to avoid repetition of similar phrases or statements within the same summary

Cold Remedy Example,
"The recommended approach to combat a cold is to drink a lot of fluids, primarily water... The advice includes consuming ample fluids, especially water... Overall, the emphasis is on hydration..."

Reduce Redundancy

Notice multiple repetitive references to drinking fluids and water, leading to redundancy

A more concise summary could be:
"The recommended approach to combat a cold is drinking plenty of fluids, especially water, along with Vitamin C supplementation for enhanced immune support."

Challenges Faced

Lack of Perspective Differentiation:

- Short inputs led to generic outputs.
- Model: The model needed detailed, varied examples to distinguish perspectives.

Prompt Conditioning Limitations:

- Simple prompts were insufficient — deeper context and structure were necessary.

Messy Dataset:

- Poor formatting and bugs in starter code increased overhead.
- Building a robust data loader took time.

Compute Constraints

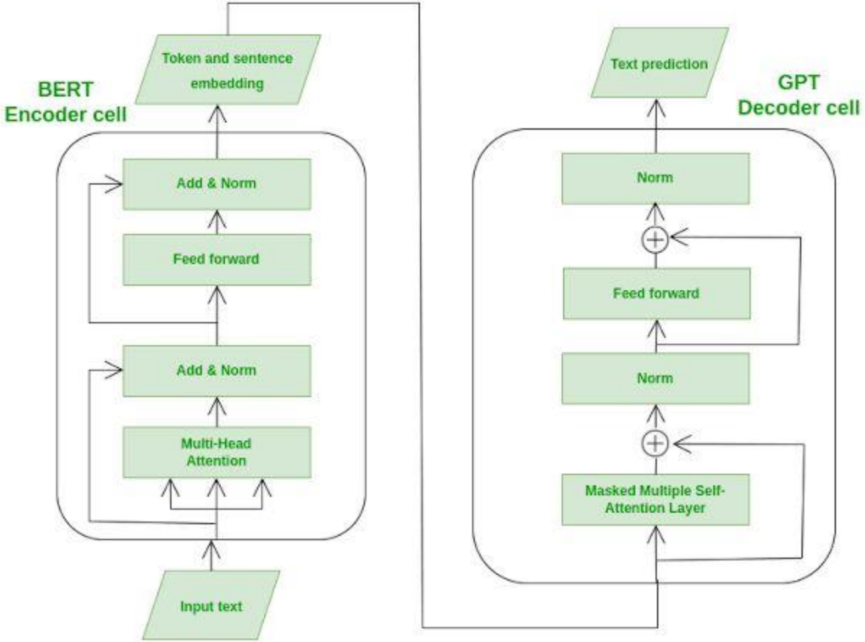
Takeaways

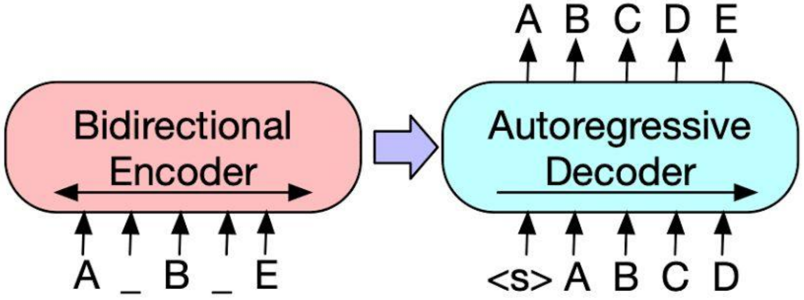
- Semantic similarity is strong (BERT-F1 ≥ 84% for all perspectives), meaning that the model generally “understands” the content but is lexical for free-form and sparse perspectives like QUESTION or EXPERIENCE.

- ROUGE-2 and BLEU remain low across all classes, indicating a scope for the requirement of more conditioning and perspective instruction; some retrieval-based/summary re-ranking method can be a fix?

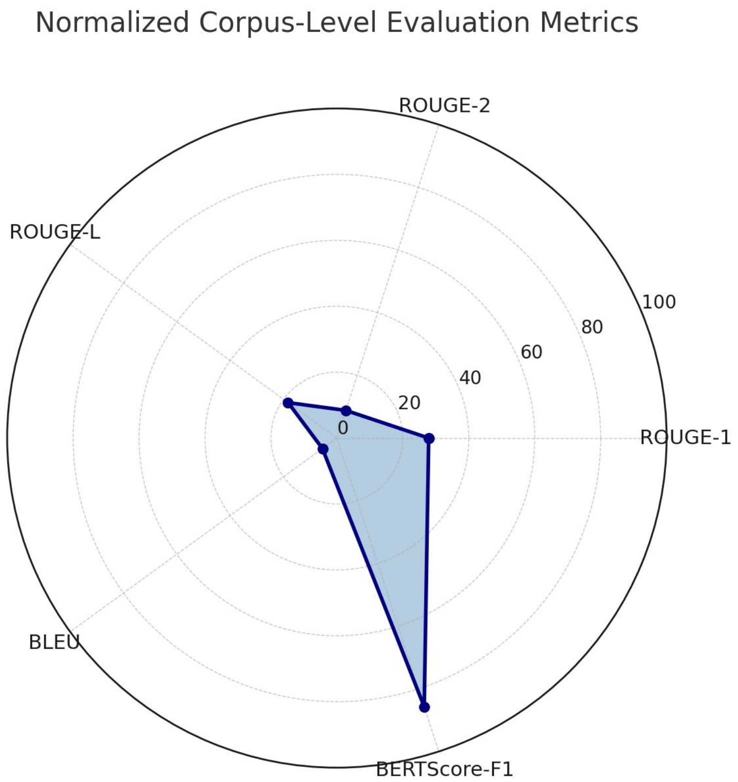
Thank You!!

Any Questions??


BERT
Encoder cell
Token and sentence embedding
Add & Norm
Feed forward
Add & Norm
Multi-Head Attention
Input text
GPT
Decoder cell
Text prediction
Norm
Feed forward
Norm
Masked Multiple Self-Attention Layer


Bidirectional Encoder
A _ B _ E
Autoregressive Decoder
A B C D E
<s> A B C D

Corpus-Level Metrics



Corpus-Level Metrics

- ROUGE-1 and ROUGE-L are moderate, showing that the summaries retain some unigram and phrase-level recall, but ROUGE-2 and BLEU are low, indicating limited n-gram overlap

- BLEU shows overall lexical similarity, relatively lower due to the abstractive nature of summaries.

- High BERTScore shows that your model generates semantically similar content, even if the surface form differs (which is common in abstractive summarization)